

D786

Microservices Design and Architecture

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A startup with 4 developers is building its first product, an appointment booking app whose features change weekly. The founders read that large streaming companies use microservices, and they have a \$20,000 cloud credit. Which architecture fits best now?

- A. A serverless function for every screen.
- B. A well-structured modular monolith, split later if clear boundaries and scaling needs appear.
- C. A separate microservice for every function, since smaller is always better.
- D. Microservices, because large streaming companies use them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An e-commerce team is splitting its monolith. Proposal 1 creates a UI service, a business-logic service, and a database service. Proposal 2 creates Orders, Payments, and Inventory services. The team hosts on a popular cloud provider. Which proposal better fits microservice design?

- A. Proposal 2, because each service owns a business capability and can change on its own.
- B. Either, as long as each runs in its own container.
- C. Whichever the cloud provider's templates suggest.
- D. Proposal 1, because layers are cleanly separated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

The Orders and Shipping services both read and write the same tables in one shared database. When the Orders team renamed a column, Shipping broke in production. The database runs on fast SSD storage. Which design change prevents this?

- A. Require both teams to approve every schema change.
- B. Move the database to faster storage.
- C. Give each service its own data store and share data only through APIs or events.
- D. Merge Orders and Shipping into one service.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Under database-per-service, the Shipping service needs a customer's address, which the Customer service owns. How should Shipping get it?

- A.** Query the Customer database directly with read-only access.
- B.** It cannot; services may never share any data.
- C.** Copy the Customer tables nightly with a database link.
- D.** Call the Customer service's API, or keep a local copy updated from its events.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A checkout request calls three services one after another, synchronously: Cart, Pricing, and Payment. Each has 99% availability. The Payment service was built by a contractor. Assuming failures are independent, what is the best-case availability of checkout?

- A.** 99.0%
- B.** About 99.9999%
- C.** About 97.0%
- D.** It depends on the contractor's track record.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An architect needs a chain of two synchronous services to reach 98.01% availability. If both services have the same availability, what must each achieve?

- A.** About 49%
- B.** 98.01%
- C.** About 99.005%
- D.** 99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A mobile app calls 7 backend services directly, each with its own address and its own login check. The app has 50,000 users in three countries. Which pattern best simplifies the client and centralizes login checks and rate limits?

- A. A service mesh.
- B. The saga pattern.
- C. A separate app build for each country.
- D. An API gateway.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

The Recommendations service has slowed to 30-second responses. Product-page requests pile up waiting on it, and the whole site slows. Recommendations was added last year and is a nice-to-have feature. Which pattern best protects the product page?

- A. Retry each slow call immediately up to five times.
- B. A circuit breaker that stops calling Recommendations after repeated timeouts and shows a fallback.
- C. Delete Recommendations because it is new.
- D. Add a cache in front of the product database.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A circuit breaker is open. After its wait period, it lets one trial request through, and the request succeeds. What state does the breaker move to?

- A. Open.
- B. Closed.
- C. Half-open.
- D. Open for another full wait period, then closed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Placing an order reserves inventory, charges the card, then books shipping, each in a separate service with its own database. Shipping fails after the card was charged. The payment provider charges a 2.9% fee. What should a saga do?

- A. Run compensating actions: refund the charge, then release the inventory.
- B. Roll back all three with one database transaction.
- C. Refund 97.1% of the charge to cover the fee.
- D. Keep the charge and retry shipping forever.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A four-step saga fails at step 3; step 3's own local transaction rolled back. Which compensations run, and in what order?

- A. None; the failed step undoes the earlier ones on its own.
- B. Steps 4, 3, 2, and 1.
- C. Step 2's, then step 1's.
- D. Step 1's, then step 2's.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A bank wants to move a 15-year-old monolith to microservices without a risky one-time cutover. The monolith is written in COBOL and must run 24/7. Which approach fits best?

- A. Route one function at a time to new services behind a facade until the monolith can be retired.
- B. Translate the COBOL into Java first, then split it.
- C. Rewrite everything and switch over in one weekend.
- D. Put an API gateway in front of the monolith and stop there.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A startup with 4 developers is building its first product, an appointment booking app whose features change weekly. ~~The founders read that large streaming companies use microservices, and they have a \$20,000 cloud credit.~~ Which architecture fits best now?

Target word: architecture fit **Decides:** Tiny team, unsettled domain.

A	Adjacent category	A hosting model, not an answer to the boundary question; still fragments a small app.
B	Correct	Microservice overhead outweighs benefits for 4 developers and a shifting domain.
C	Absolute language	Too-fine services add network and operational cost with no benefit here.
D	Camouflage	Other firms' scale does not describe this team's needs.

2. Answer A Concept 2

An e-commerce team is splitting its monolith. Proposal 1 creates a UI service, a business-logic service, and a database service. Proposal 2 creates Orders, Payments, and Inventory services. ~~The team hosts on a popular cloud provider.~~ Which proposal better fits microservice design?

Target word: choose service boundaries **Decides:** Business capability vs. technical layer.

A	Correct	Capability boundaries keep most changes inside one service.
B	Dropped element	Containers do not fix poor boundaries.
C	Camouflage	The host does not decide domain boundaries.
D	Confusion set	Layer splits make almost every feature touch all three services.

3. Answer C Concept 3

The Orders and Shipping services both read and write the same tables in one shared database. When the Orders team renamed a column, Shipping broke in production. ~~The database runs on fast SSD storage.~~ Which design change prevents this?

Target word: remove data coupling **Decides:** Two services depend on one schema.

A	Adjacent category	Adds coordination but keeps the coupling.
B	Camouflage	Speed was not the problem.
C	Correct	A service can change its schema without breaking others.
D	Inverted operation	Moves back toward the monolith instead of decoupling.

4. Answer D Concept 3 · transfer

Under database-per-service, the Shipping service needs a customer's address, which the Customer service owns. How should Shipping get it?

Target word: legal way to reach another service's data **Decides:** Data is reached only through the owner.

A	Confusion set	Read-only still couples Shipping to Customer's schema.
B	Absolute language	They share data, just through the owning service.
C	Adjacent category	Still reads the other schema directly.
D	Correct	The owner controls access, so its schema can change safely.

5. Answer C Concept 4

A checkout request calls three services one after another, synchronously: Cart, Pricing, and Payment. Each has 99% availability. ~~The Payment service was built by a contractor.~~ Assuming failures are independent, what is the best-case availability of checkout?

Target word: chain availability **Decides:** Multiply availabilities in a chain.

A	Dropped element	Drops the effect of chaining three services.
B	Inverted operation	Treats them as redundant backups instead of a chain.
C	Correct	$0.99 \times 0.99 \times 0.99 = 0.9703$.
D	Camouflage	The stated 99% already reflects that service's availability.

6. Answer D Concept 4 · transfer

An architect needs a chain of two synchronous services to reach 98.01% availability. If both services have the same availability, what must each achieve?

Target word: solve for per-service availability **Decides:** $a \times a = 0.9801$.

A	Inverted operation	Divides by two instead of taking the square root.
B	Wrong stem number	The chain target, which each link must exceed.
C	Confusion set	Averages the target with 100%, which is not how chains combine.
D	Correct	$\sqrt{0.9801} = 0.99$.

7. Answer D Concept 5

A mobile app calls 7 backend services directly, each with its own address and its own login check. ~~The app has 50,000 users in three countries.~~ Which pattern best simplifies the client and centralizes login checks and rate limits?

Target word: single entry point pattern **Decides:** Client-to-backend calls need one front door.

A	Adjacent category	Manages service-to-service traffic, not the client's entry.
B	Confusion set	Coordinates distributed transactions, not client routing.
C	Camouflage	Geography does not reduce the 7 direct calls.
D	Correct	One entry point routes requests and handles cross-cutting concerns.

8. Answer B Concept 6

The Recommendations service has slowed to 30-second responses. Product-page requests pile up waiting on it, and the whole site slows. ~~Recommendations was added last year~~ and is a nice-to-have feature. Which pattern best protects the product page?

Target word: protect callers from a slow dependency **Decides:** A failing, optional dependency is dragging down callers.

A	Inverted operation	Adds load to the struggling service and waits longer.
B	Correct	Fails fast so product-page threads are freed.
C	Camouflage	Its age does not decide the pattern.
D	Adjacent category	Speeds a different dependency; the slow call remains.

9. Answer B Concept 6 · transfer

A circuit breaker is open. After its wait period, it lets one trial request through, and the request succeeds. What state does the breaker move to?

Target word: state after a successful trial **Decides:** Success in half-open closes the breaker.

A	Inverted operation	Open follows a failed trial, not a success.
B	Correct	A successful trial means normal traffic resumes.
C	Confusion set	That is the trial state it is leaving.
D	Dropped element	Ignores the successful trial.

10. Answer A Concept 7

Placing an order reserves inventory, charges the card, then books shipping, each in a separate service with its own database. Shipping fails after the card was charged. ~~The payment provider charges a 2.9% fee.~~ What should a saga do?

Target word: saga failure handling **Decides:** Earlier steps already committed locally.

A	Correct	Compensations undo completed steps in reverse order.
B	Confusion set	Separate databases cannot share one local rollback.
C	Camouflage	The fee is not what the pattern decides.
D	Absolute language	Never ending retries leave the customer charged with no order.

11. Answer C Concept 7 · transfer

A four-step saga fails at step 3; step 3's own local transaction rolled back. Which compensations run, and in what order?

Target word: which steps to compensate **Decides:** Only completed steps are undone, newest first.

A	Dropped element	Drops the compensations the saga must trigger.
B	Wrong stem number	Step 4 never ran and step 3 already rolled back.
C	Correct	Steps 1 and 2 completed; undo in reverse order.
D	Inverted operation	Wrong order; the newest completed step is undone first.

12. Answer A Concept 8

A bank wants to move a 15-year-old monolith to microservices without a risky one-time cutover. ~~The monolith is written in COBOL~~ and must run 24/7. Which approach fits best?

Target word: migration approach **Decides:** Must avoid a big-bang cutover.

A	Correct	The strangler fig pattern migrates gradually with low risk.
B	Camouflage	Language is not the migration strategy.
C	Inverted operation	That is the big-bang cutover the bank wants to avoid.
D	Dropped element	A facade alone moves nothing out of the monolith.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	When microservices fit	Microservices pay off when many teams need to deploy and scale parts independently; a small team with a changing domain is usually better served by a modular monolith.	Choosing microservices because large companies use them.	Team size and how settled the domain is.
2	Boundaries by business capability	Split services around business capabilities (orders, payments), not technical layers (UI, logic, database).	Splitting by layer, which forces every change through every service.	Two proposed splits, one by layer and one by business function.
3	Database per service	Each service owns its data; others reach it only through the service's API or events.	Sharing tables between services for convenience.	One service's schema change breaks another.
4	Synchronous call chains and availability	Services called in a synchronous chain multiply availabilities, so the chain is less available than any single link.	Assuming the chain is as available as each service.	Several services called in sequence, each with a percentage.
5	API gateway	A single entry point routes client requests and centralizes cross-cutting concerns such as authentication and rate limiting.	Confusing it with a service mesh, which handles service-to-service traffic.	Clients calling many backend addresses directly.
6	Circuit breaker	After repeated failures, stop calling the failing service and use a fallback; half-open tests recovery, success closes the breaker.	Retrying harder against a struggling service.	A slow dependency dragging down its callers.
7	Saga pattern	A multi-service transaction runs as local steps; on failure, compensating actions undo completed steps in reverse order.	Expecting one database rollback across services.	A later step fails after earlier services already committed.
8	Strangler fig migration	Replace a monolith gradually by routing one function at a time to new services behind a facade until the old system can be retired.	Big-bang rewrites and cutovers.	A risky legacy system that must keep running.

Course scope (public description): In this course, students will demonstrate their knowledge of microservice design and architecture. Students will analyze a business's needs to find what best matches their software architecture needs. Additionally, students will follow key design principles to create a microservice while explaining why the design pattern is the most efficient for a project. Upon completion of the course, students will create a deployment strategy for efficiency and effectiveness.